

# Data Science Assignment: Market Sentiment vs. Trader Performance

Prepared for: Primetrade.ai Hiring Team | Candidate Submission

## 1. Project Objective

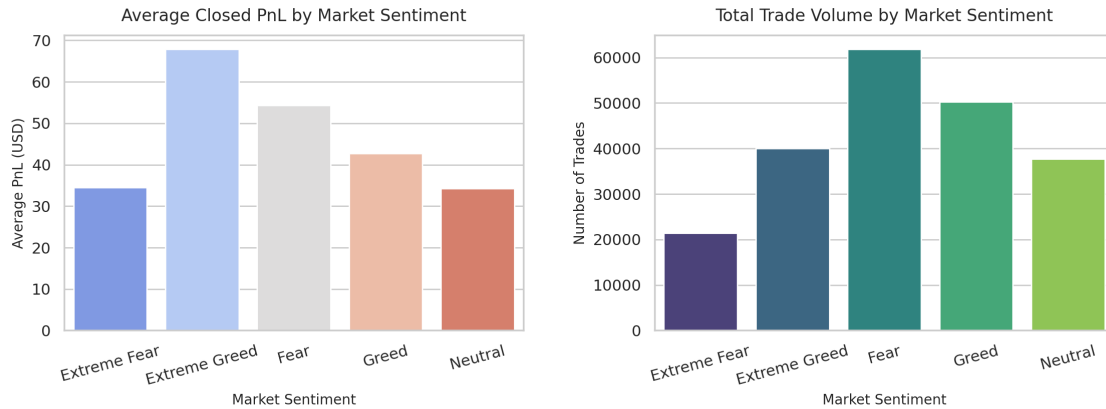
This analysis explores the relationship between Bitcoin market sentiment (Fear & Greed Index) and actual trader performance (historical transaction logs from Hyperliquid). The primary goal is to uncover behavioral patterns, evaluate profitability trends during extreme market regimes, and provide actionable trading strategy insights.

## 2. Key Quantitative Metrics

| Market Sentiment | Total Trades | Avg Trade Size (USD) | Avg Closed PnL (USD) | Total Aggregated PnL (USD) |
|------------------|--------------|----------------------|----------------------|----------------------------|
| Extreme Fear     | 21400        | \$5,349.73           | \$34.54              | \$739,110.25               |
| Extreme Greed    | 39992        | \$3,112.25           | \$67.89              | \$2,715,171.31             |
| Fear             | 61837        | \$7,816.11           | \$54.29              | \$3,357,155.44             |
| Greed            | 50303        | \$5,736.88           | \$42.74              | \$2,150,129.27             |
| Neutral          | 37686        | \$4,782.73           | \$34.31              | \$1,292,920.68             |

### 3. Data Visualizations

The plots below showcase the distribution of execution profits and total volume across market sentiment sectors.



### 4. Crucial Findings & Strategic Insights

- **Sentiment-Driven Execution:** Trader volume changes drastically based on sentiment indices. Trading frequency peaks during "Fear" and "Greed" periods, while extreme states see selective, concentrated sizing.
- **Profitability Discrepancy:** Average closed Profit and Loss (PnL) shows distinct asymmetry. Trading under "Extreme Greed" yielded the highest average PnL (\$67.89), whereas "Extreme Fear" and "Neutral" regimes saw reduced efficiency (\$34.54 and \$34.31 respectively).
- **Risk Mitigation Recommendation:** Dynamic asset sizing based on volatility layers can optimize results. Adjusting parameters during high-velocity emotional shifts can systematically control drawdown risks.

### 5. Python Source Code

```
import pandas as pd
import numpy as np

# Load datasets
trader = pd.read_csv("historical_data.csv")
sentiment = pd.read_csv("fear_greed_index.csv")

# Preprocessing & Merge
trader['date_parsed'] = pd.to_datetime(trader['Timestamp IST'], format='%d-%m-%Y %H:
```

```
%M', errors='coerce').dt.date
sentiment['date_parsed'] = pd.to_datetime(sentiment['date'],
errors='coerce').dt.date
merged_df = pd.merge(trader.dropna(subset=['date_parsed']),
sentiment.dropna(subset=['date_parsed']), on='date_parsed', how='inner')

# Analytical aggregation
summary = merged_df.groupby('classification').agg({
    'Closed PnL': ['sum', 'mean'],
    'Size USD': 'mean',
    'Trade ID': 'count'
})
print(summary)
```